Development of a New Device to Address Racial Bias in Pulse Oximetry

Akshat Jain[1], Rohan Bendapudi[1], Kuhu Jayaswal[1], Chhavi Goenka[1]
Olin College of Engineering, Health Equity and Access Lab, Needham, MA, 02494

INTRODUCTION

Pulse oximeters overestimate the oxygen saturation in people with higher melanin concentration limiting their ability to get care or coverage. Our research has identified the sources of error to be the relationship between oxygen saturation and red-to-infrared ratio and insufficient information about the contribution of skin pigmentation to the measurements. Our design addresses sources of errors due to the racial inequities in data from existing clinical trials, we initially used precisely designed optical phantoms, followed by in vitro tests.

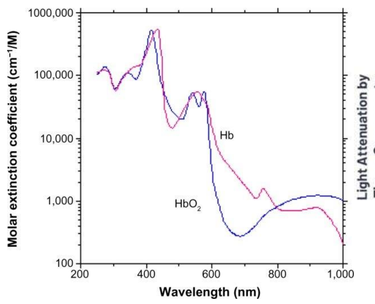


Figure 1: Graph shows the molar extinction coefficient of hemoglobin at a certain wavelength

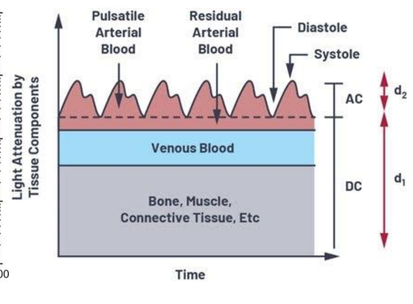


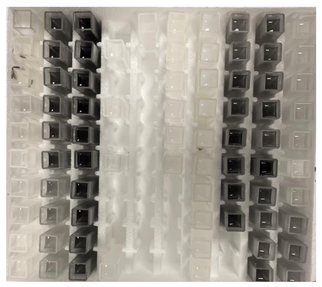
Figure 2: SpO2 level measurement for different concentrations of oxygenated hemoglobin and reduced hemoglobin by measuring the PPG signal at two different wavelengths.

BACKGROUND

- Pulse oximeters measure arterial blood oxygen saturation, percentage of hemoglobin statured with oxygen, providing continuous, non-invasive data on hemoglobin oxygenation. Widely used, they can't be user-calibrated for practical purposes.
- The existing bias in pulse oximeters lead to inaccurate oxygen saturation readings in individuals with higher melanin content, posing significant challenges such as hindering timely hospital admissions and impeding access to insurance coverage for essential oxygen support. One possible explanation for this discrepancy is the lack of racial diversity in the original population used in the development of pulse oximeters.
- We have determined three possible causes of error due to racial bias in pulse oximeters
    1. The relationship between oxygen saturation and ratios of red and infrared intensities that was determined by trials on a mostly white population
    2. Lack in the incorporation of wavelengths that determine the contribution of melanin
    3. Insufficient information about the contribution of skin pigmentation to the DC component of the pulse oximeter measurements

METHODOLOGY

- We are using optical phantoms to simulate varying melanin concentrations for non-invasive testing. Our design addresses the sources of errors using three methods - addition of a wavelength, considering the contribution of the DC signal in the pulse oximeter measurements and designing new algorithms to calculate the oxygen saturation.
- In our circuit, we employ LEDs of multiple wavelengths, a photodiode for detection, and a spectrometer for validation, with all acquired data processed and analyzed in MATLAB.



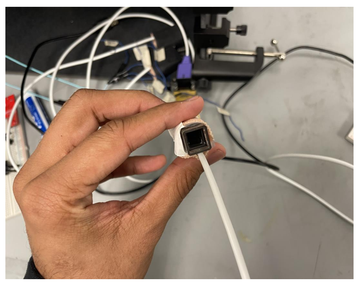


Figure 3: Optical phantoms used to simulate different melanin concentrations, phantoms were mainly built using different concentrations of India ink and milk.

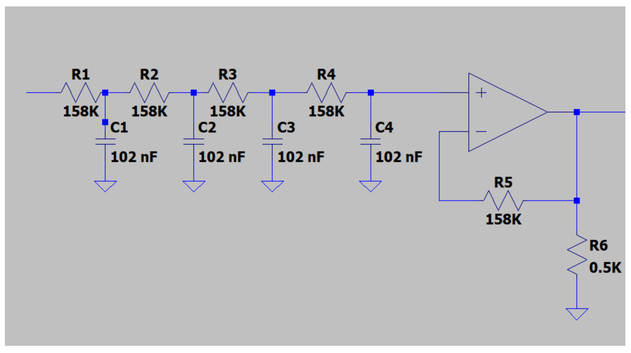


Figure 4: Stage 1. We designed a 10 Hz fourth-order low-pass filter, applied Fourier transform techniques to pinpoint a more optimal cutoff frequency for noise reduction, and then amplified the output signal by 4 for enhanced clarity and strength.

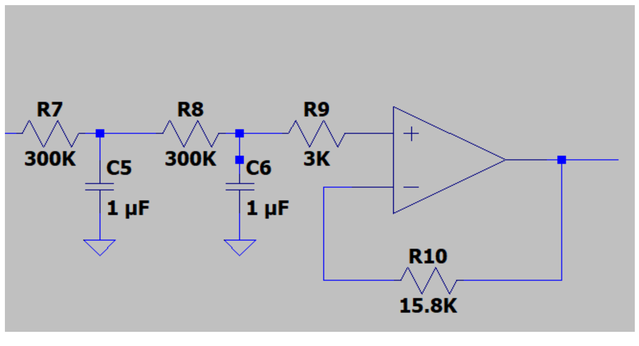


Figure 5: Stage 2 To isolate the DC component from an AC signal, we implemented a second-order low-pass filter with a 0.5 Hz cutoff frequency, effectively separating the two. Afterward, we amplified both signals by a factor of 5 and created parallel signal channels for independent processing of the DC and AC outputs.

RESULTS & ONGOING WORK

- Our design is a three-pronged approach that addresses these sources of error A new algorithm to calculate oxygen saturation from the pulse oximeter measurements
- Additional wavelengths that will account for the contribution of melanin to the data
- Design of control electronics that better determine the DC contribution of melanin to the measurements

The challenges we face are lack of data due to the racial inequities in existing clinical trials. To circumvent this, we are initially using precisely designed optical phantoms.

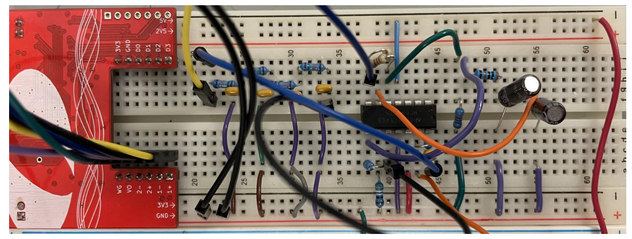


Figure 6: Circuit diagram of stage 1 and 2 with low pass filters and isolating the DC component of the signal.

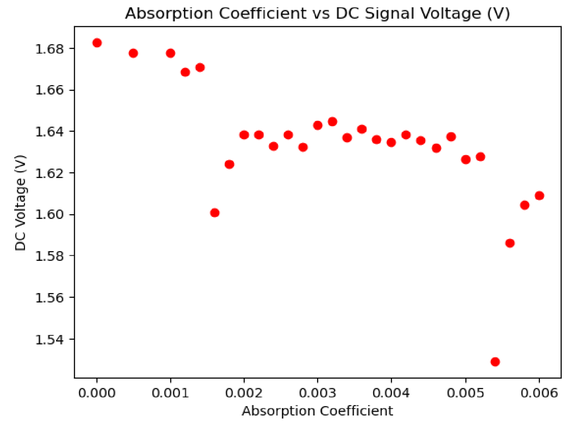


Figure 7: A scatter plot highlighting the absorption coefficient against the filtered DC voltage from the pulse oximeter

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